



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

INTERNET OF THINGS (IoT) AND SMART SYSTEMS

Connecting Devices, Data And Intelligence

CSA1003-SOFTWARE ENGINEERING

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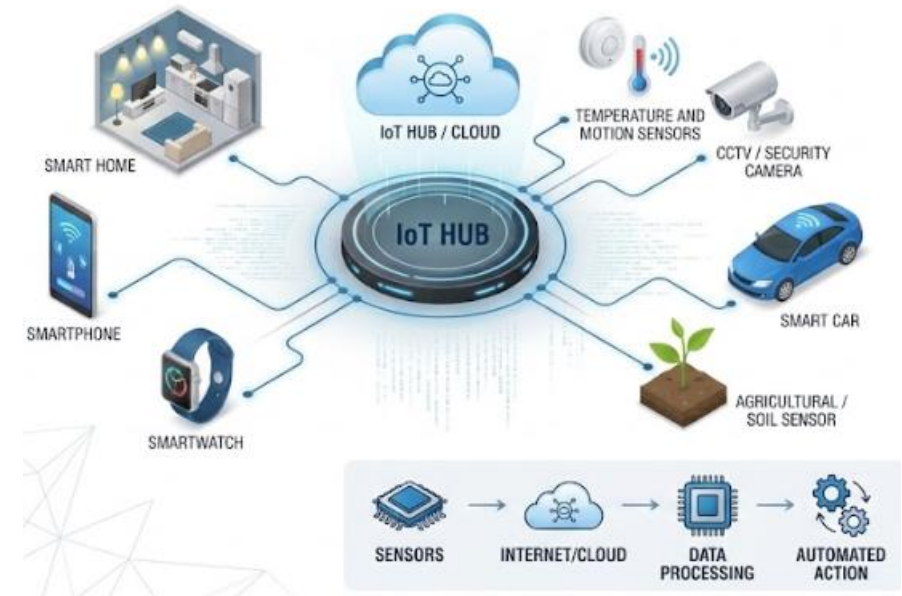
Introduction

What is IoT?

- IoT stands for **Internet of Things**.
- It connects physical devices to the Internet.
- Devices collect, exchange, and process data.
- Sensors and actuators enable interaction with the physical world.
- IoT enables **automation, monitoring, and remote control**.

What are Smart Systems?

- Systems that use sensors, connectivity, and intelligent processing.
- They can monitor conditions and make automated decisions.



Need and Motivation

Why is IoT Needed?

- Increasing demand for automation
- Real-time monitoring of systems
- Reduces human effort
- Improves efficiency and productivity
- Enables remote access and control
- Helps in better decision-making
- Supports efficient use of resources





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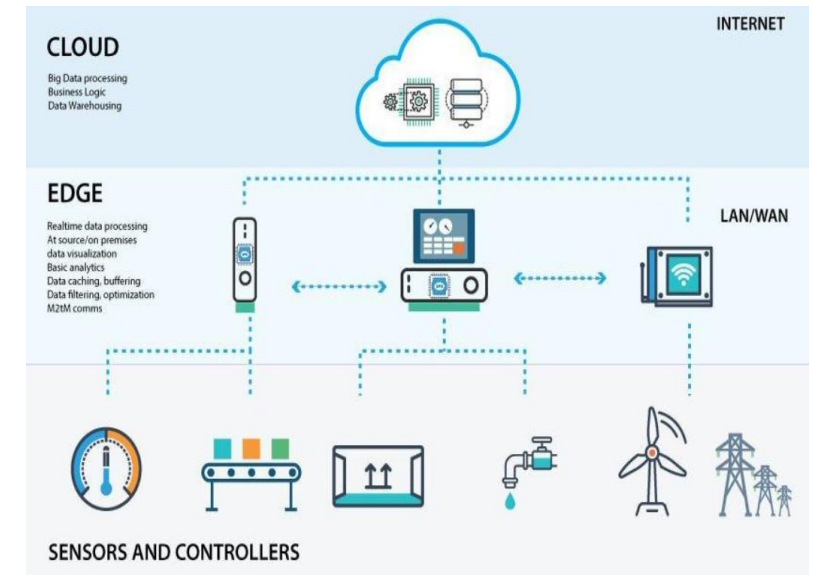
Working Principle

How IoT Works

- **Sensing** – Sensors collect data.
- **Connectivity** – Data is transmitted through a network.
- **Processing** – Cloud or edge systems process the data.
- **Analysis** – Data is analyzed to identify useful information.
- **Decision** – The system decides what action is required.
- **Action** – Actuators perform the required operation.

Basic Flow:

- **Sensors → Network → Cloud/Edge → Processing → Decision → Actuator**

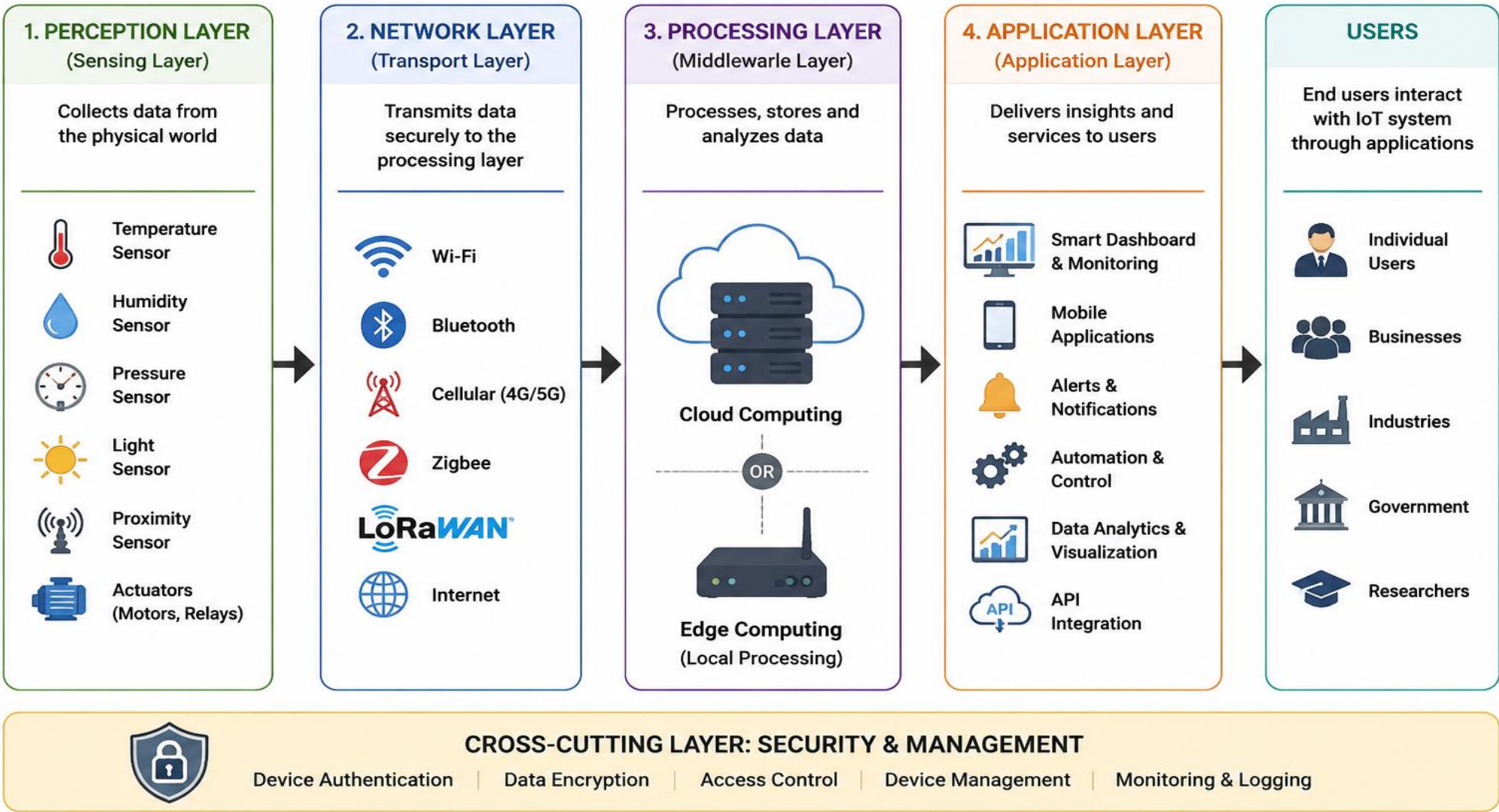




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IoT Architecture





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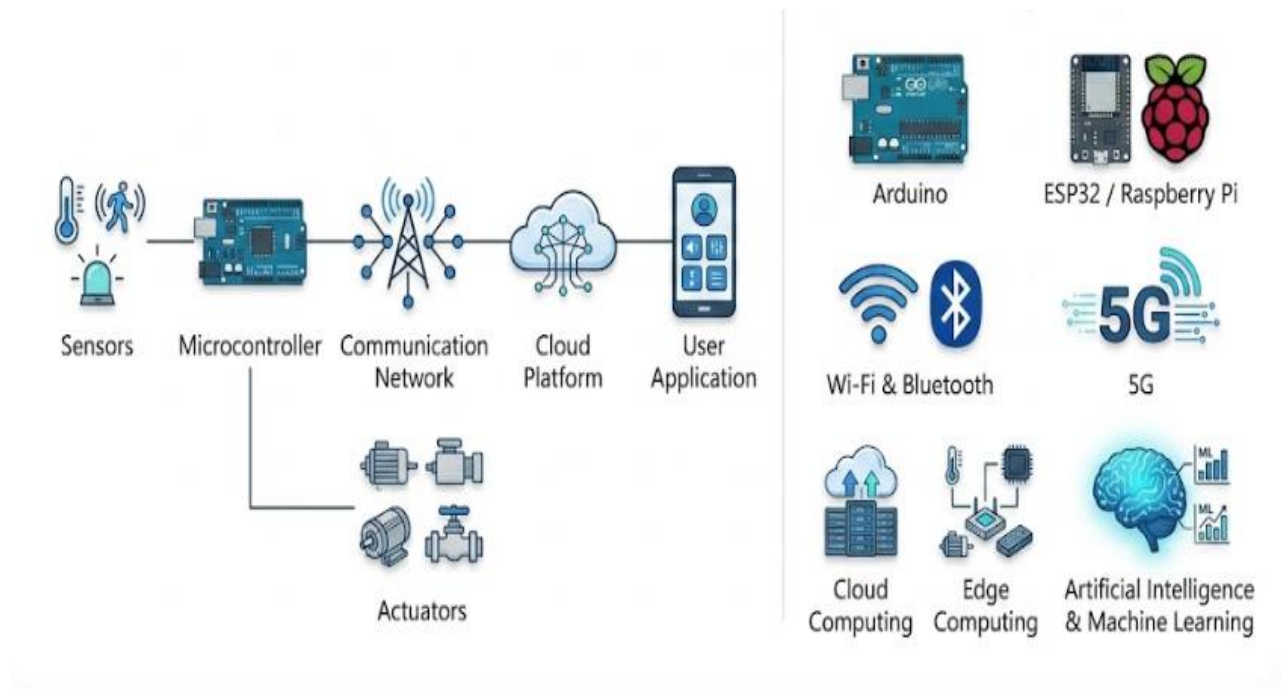
Key Components & Technologies

- **Key Components**

- Sensors
- Actuators
- Microcontrollers
- Communication networks
- Cloud platforms
- User applications

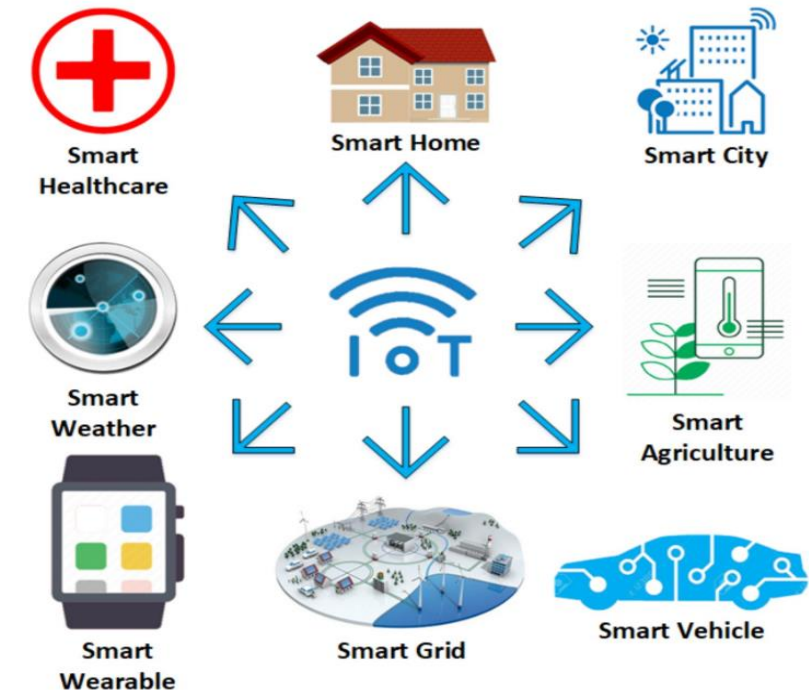
- **Technologies Used**

- Arduino
- ESP32 / Raspberry Pi
- Wi-Fi & Bluetooth
- 5G
- Cloud Computing
- Edge Computing
- Artificial Intelligence & Machine Learning



Applications of IoT

- **Smart Homes** – Lighting, security & temperature control
- **Smart Healthcare** – Patient monitoring & wearables
- **Smart Agriculture** – Soil monitoring & smart irrigation
- **Smart Industries** – Machine monitoring & predictive maintenance
- **Smart Cities** – Traffic, parking & waste management



Advantages & Limitations

Advantages

- Automation
- Real-time monitoring
- Improved efficiency
- Reduced cost
- Remote control

Limitations

- Security risks
- Privacy issues
- Internet dependency
- High initial cost
- Compatibility issues

Advantages & Limitations of IoT

Advantages

- ✓  Automation
- ✓  Real-time Monitoring
- ✓  Improved Efficiency
- ✓  Reduced Cost
- ✓  Remote Control

Limitations

- ✗  Security Risks
- ✗  Privacy Issues
- ✗  Internet Dependency
- ✗  High Initial Cost
- ✗  Compatibility Issues



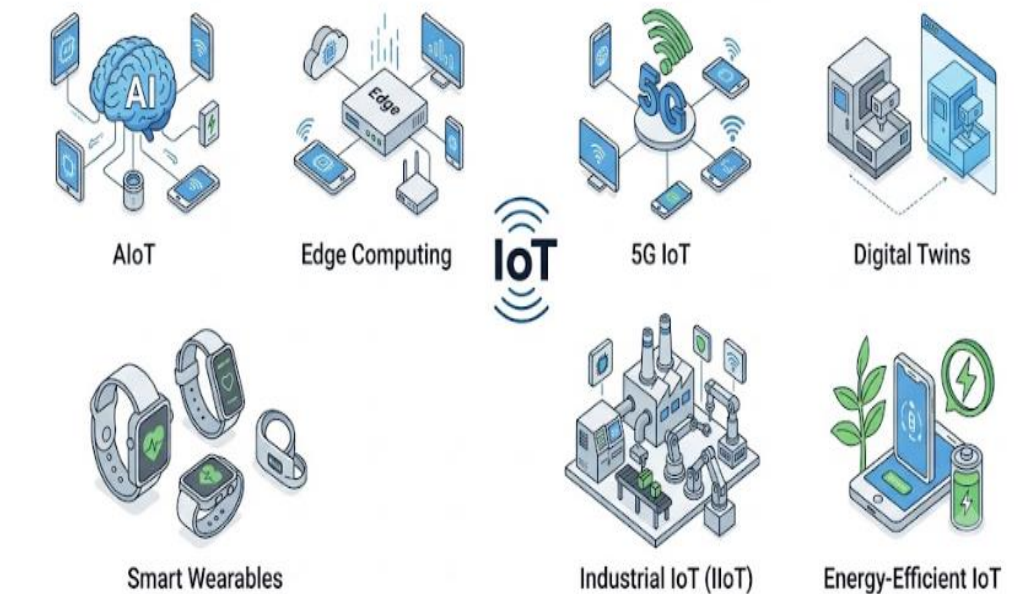
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Recent Developments & Trends

Emerging Trends in IoT

- **AIoT** – Integration of AI with IoT
- **Edge Computing** – Processing data closer to devices
- **5G IoT** – Faster and reliable connectivity
- **Digital Twins** – Virtual representation of physical systems
- **Smart Wearables**
- **Industrial IoT (IIoT)**
- **Energy-efficient IoT devices**





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Challenges & Future Scope



Challenges

- Cybersecurity
- Data privacy
- Scalability
- Standardization
- Power consumption
- Reliable connectivity

Future Scope

- Autonomous smart systems
- AI-powered IoT
- Intelligent healthcare
- Smart transportation
- Sustainable smart cities
- Fully automated industries



Cybersecurity



Data Privacy



Scalability



Standardization



Power Consumption



Reliable Connectivity



Autonomous Smart Systems



AI-Powered IoT



Intelligent Healthcare



Autonomous cars



Smart Transportation



Smart Cities



Sustainable Smart Cities



Fully Automated Industries



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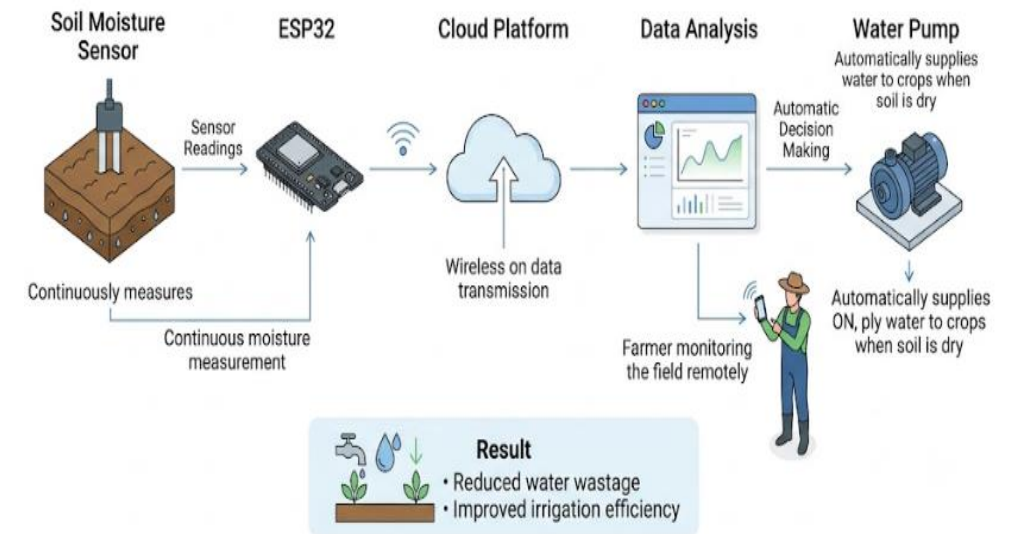
Case Study: IoT-Based Smart Agriculture

Working:

- **Soil Moisture Sensor → ESP32 → Cloud → Data Analysis → Water Pump**
- Soil moisture is continuously measured.
- ESP32 collects sensor readings.
- Data is sent to the cloud.
- System analyzes soil conditions.
- Water pump automatically turns ON when required.
- Farmer can monitor the field remotely.

Result:

Reduces water wastage and improves irrigation efficiency.





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Conclusion



- IoT connects physical devices and enables data exchange.
- Smart systems use IoT data to monitor and control processes intelligently.
- IoT improves automation, efficiency, and decision-making.
- Integration with **AI, Edge Computing, Cloud Computing, and 5G** is making IoT more powerful.
- IoT will play a major role in building a **smarter and more connected future**.



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References



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*Thank
you!*